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The C-Factor Series

A mechanism-first review of the sciences that underlie medicine - built for the MCAT and as a running start on USMLE Step 1.

What this is

Seven connected books that teach the premedical sciences as one story. The series does not hand you lists to memorize; it teaches the mechanism beneath each fact, so an unfamiliar exam question becomes something you can reason through. Each book is a single arc with a shared box system: discriminators (look-alikes you must tell apart), clinical bridges (where a concept becomes a disease, drug, or decision), why-boxes (the physical reason a thing is true), and high-yield anchors (the must-keep facts).

Who it is for

- Premed students preparing for the MCAT who want understanding, not flashcard cramming.
- Incoming and early medical students who want a mechanism-first running start on Step 1 content.
- Anyone re-building science fundamentals (biology, chemistry, organic, physics, psych/soc, biochemistry) from the ground up.

What is included

Component	What it is
7 Books	~2,270 pages, mechanism-first, cover on page 1
92 Lecture decks	One PowerPoint per chapter, with presenter teaching notes
6,199 Flashcards	Anki decks (.apkg) from every discriminator, bridge, why, and anchor
Cheat sheets	One-page rapid-review PDF per chapter
Study syllabus	Mapped 8-week and 12-week plans

The booklist

#	Title	Subject
1	The Living Code	Biology
2	First Principles	General Chemistry
3	Carbon's Architecture	Organic Chemistry
4	Forces and Flow	Physics
5	Mind and Society	Psychology & Sociology
6	The Molecular Story	Biochemistry
7	The Final Review	All-subject high-yield

How to use it - the three passes

- Pass 1 - Mechanism.** Read for the why; watch the discriminator and why boxes.
- Pass 2 - Clinical + questions.** Work clinical bridges and passages; start Anki.
- Pass 3 - Rapid review.** Cheat sheets + high-yield anchors; Anki carries retention.

See the Study Syllabus for the mapped 8-week and 12-week schedules.

A taste of the method

Below are real samples pulled straight from the books, so you can see how each tool reads.

Sample discriminator

ALPHA vs BETA ANOMERS - WHICH IS MORE STABLE?

For D-glucose specifically: the BETA anomer is more stable in solution (approximately 64% at equilibrium) because the C1 hydroxyl occupies an EQUATORIAL position in the chair conformation, minimizing 1,3-diaxial interactions with the C3 and C5 ring substituents. The ALPHA anomer places the C1 hydroxyl AXIAL, which is less stable due to steric crowding. Memory anchor: Alpha = Axial = less stable for glucose. However, this stability preference reverses for fructose in some conformations - a classic MCAT trap. More importantly, the alpha vs beta distinction governs glycosidic bond type and therefore digestibility: alpha-1,4 bonds in starch are cleaved by human amylases; beta-1,4 bonds in cellulose are not, because humans lack beta-glucosidase (cellulase). The anomeric configuration of glycosidic bonds is therefore not a trivial structural detail but a functionally decisive one.

Sample why-box

ATP IS IN THE MIDDLE OF THE ENERGY HIERARCHY

The phosphate group transfer potential of a compound determines whether it can spontaneously donate phosphate to another compound. Compounds with high phosphate group transfer potential (PEP, 1,3-BPG, creatine phosphate) can donate phosphate to ADP, producing ATP. ATP in turn can donate phosphate to compounds with lower transfer potential (glucose, glycerol, proteins). If ATP were the highest-energy compound, nothing could make it. If ATP were the lowest-energy compound, it could not power reactions. By sitting in the middle of the hierarchy, ATP can be made from the high-energy intermediates of catabolism and spent to drive the low-energy intermediates of anabolism. This is the design logic of the energy currency: intermediate position enables the two-way role of donor and acceptor. Creatine phosphate serves as a rapidly accessible phosphate reserve in muscle because its hydrolysis ΔG° (-43 kJ/mol) is negative enough to drive ATP synthesis from ADP.

Sample high-yield anchors

- Alpha anomeric OH = axial (less stable for glucose); beta = equatorial (more stable; ~64% at equilibrium)
- D/L: configuration at the chiral center FARTHEST from carbonyl; almost all biological sugars are D
- Epimers: glucose/galactose differ at C4; glucose/mannose differ at C2
- Reducing sugar = free anomeric carbon; sucrose is NOT reducing (both anomeric carbons locked in bond)
- Beta-1,4 bonds (cellulose, chitin) are indigestible by mammals - no cellulase
- Glycogen: more branches than amylopectin (every 8-12 vs 24-30 residues) for faster glucose release

Branding is Bit by Bit MD - The C-Factor Series. Mechanism first. Memorization second.